

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I N. Sai yashaswini ¹⁹²⁴²⁵²⁵⁰ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Scenario 2: Intercepted suspicious Message

Given ciphertext

GSRH RH ZHVXING

1. Most Likely Message (original)

The original message is

THIS IS A SECRET

The encryption technique used in Atbash cipher, which is a simple substitution cipher

In Atbash, the Alphabet is Reversed

Normal: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Cipher: Z Y X W V U T S R Q P O N M L K J I H G F E D C B A

Therefore: G → T

S → H

R → I

H → S

R → I

H → S

Z → A

H → S

V → E

X → C

I → R

V → E

G → T

Hence

GSRH RH Z HVXING

THIS IS A SECRET

2. Justification:

The word structure of the ciphertext is

4-2-1-6

GSRH RH Z HVXING

The decrypted text has the same structure

THIS IS A SECRET

Result:

Ciphertext: GSRH RH Z HVXING

Plaintext: THIS IS A SECRET

Technique: Atbash Substitution Cipher

Scenario 5: Legacy Encryption in Corporate Communication

Given Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

The problem states that the encryption uses three levels of arrangement. This indicates a 3-Rail Rail Fence Cipher, which is a transposition cipher.

1. Most probable Original Message:

WE ARE DISCOVERED FLEE AT ONCE

without spaces:

WEAREDISCOVEREDFLEEATONCE

In a 3-Rail Rail fence Cipher, the message is written in a zig-zag pattern across 3 rows.

Rail 1: W E D S O E L E T N

Rail 2: E R I C V R D F E A O C

Rail 3: A S E E F A C

Reading the rows gives

WECRLTEERDSOEFEAOCAIVDEN

Thus reversing the process gives

WE ARE DISCOVERED FLEE AT ONCE

2. why is the Method vulnerable?

The Real Rail Fence cipher is considered insecure for Modern communication for several reasons.

- a) simple Rearrangement
- b) Easy to guess
- c) Easy for computers to break
- d) weak Confidentiality
- e) NO integrity protection.

Therefore the Rail Fence cipher should not be used for sensitive corporate communication

3. practical Enhancement:

The company should replace the legacy Rail Fence Cipher with a modern encryption Algorithm such as AES-256-GCM

Advantages:

- provides strong confidentiality
- use a 256-bit key
- Detects unauthorized modification
- Suitable for modern communication systems.
- provides both encryption and authentication

The improved communication is

Original Message



AES-256 GCM Encryption



Encrypted Message



Secure communication



AES-256-GCM → Decryption → Original Message

The encryption keys should ^{also} be stored and exchanged using a secure key-management system.